

Ayush Kumar

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EDUCATION

Manipal University Jaipur

Bachelor of Technology in Computer Science (Data Science)

Jaipur, Rajasthan

2023 – 2027

- **CGPA:** 8.25 / 10.0

DAV Public School

CBSE Class XII (PCM)

Ranchi, Jharkhand

2022 – 2023

- **Percentage:** 84.6%

EXPERIENCE

Software Development Engineer (SDE) Intern

BHRM AI

Jan. 2026 – Feb. 2026

Jaipur, Rajasthan (Remote)

- Engineered a high-performance Backend Resource Management Hub (BRMH) using **Node.js** and **AWS**, supporting complex resource orchestration across multiple namespaces.
- Developed an advanced caching layer using **AWS ElastiCache (Valkey)** and **Redis**, implementing non-blocking background updates and parallel configuration processing to reduce API latency by 60%.
- Integrated **LangChain** and **Anthropic (Claude 3.5 Sonnet)** to build an intelligent AI agent capable of automated web scraping, schema generation, and serverless **Lambda** function deployment.
- Architected a robust **RBAC** and **SSO** system using **AWS Cognito**, ensuring secure multitenant access and fine-grained permission control.
- Designed a scalable workflow engine supporting both **HTTP** and **Lambda** execution paths, enabling automated data synchronization and API migration.
- Optimized database operations on **DynamoDB** with sequential chunking and bounded buffers, handling large-scale datasets while maintaining sub-second response times.

PROJECTS

TypeSync | *Socket.io, Node.js, React, Express, MongoDB*

- Built a real-time multiplayer typing platform using **Socket.io** for instant data synchronization.
- Designed scalable **Node.js** backend services for room management and live score updates.
- Implemented real-time WPM and accuracy tracking with sub-second latency and managed high-frequency DOM updates using **React's useRef** to avoid unnecessary re-renders.
- Developed a robust Room Management system supporting private invites via unique UUID-based URLs and public matchmaking queues.

PoPo Podcast | *Next.js, WebRTC, FFmpeg, Node.js, React*

- Engineered a real-time audio/video streaming platform using **WebRTC** and **FFmpeg** for low-latency media delivery.
- Leveraged **Next.js** Server-Side Rendering (SSR) and Incremental Static Regeneration (ISR) to improve SEO and optimized media pipelines to support concurrent users.
- Implemented client-side stream recording using the **MediaRecorder API**, allowing users to capture high-quality WebRTC streams directly in the browser with zero server-side lag.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Java, C++, Python, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend & Cloud: Node.js, Express.js, Socket.io, WebRTC, AWS (Lambda, DynamoDB, Cognito, S3, EventBridge, IAM)

Databases & Tools: MongoDB, Redis, Git, GitHub, Postman, FFmpeg, Docker

Machine Learning: Credit Card Fraud Detection using Random Forest, feature engineering, class imbalance handling